Forgetting to Refuse: Does a Taught Safeguard Survive Benign Personalization of a Vision-Language-Action Policy?

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Abstract: Benign fine-tuning degrades refusal in aligned language models, but whether an *action-level* safeguard in a vision-language-action (VLA) policy survives benign personalization has not been measured. We teach OpenVLA-7B to hold still under harmful instructions (supervised LoRA on paired scene states), personalize it on 200 benign demonstrations of other tasks, and re-measure refusal, task success and closed-loop contact with a static human hand on identical states, against a comparator never taught the safeguard. Two findings hold up. First, the physical benefit of alignment did not survive: before personalization the aligned policy contacted the hand in 2% of benign-instruction episodes against the comparator's 72%; afterwards it was no safer than the comparator under benign or blank instructions (16% vs. 10%, 28% vs. 24%), and an exploratory extension to 12 epochs left them equal. Second, the first-step refusal rate that text-model safety work would carry over is not specific: it fell in all three alignment seeds (mean $0.90 \rightarrow 0.59$) but also *rose* in the comparator, which was never taught to refuse. The pre-registered primary outcome, harmful-instruction contact, was null (0/50 before and after), but no policy in the study, aligned or not, executed the harmful clause, so this outcome cannot register compliance. Closed-loop harm metrics for VLA refusal need an arm that demonstrably complies.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], standard visual instruction tuning makes vision-language models less safe than their language backbones [2], and alignment that shapes only the first output tokens is especially fragile [3, 4]. All of these results concern text or image outputs. VLA policies produce actions, and safety is now being installed in their weights [5] and measured with closed-loop benchmarks that count physical violations [6], but each such safeguard is trained once and evaluated once. A deployed VLA is instead personalized on a few dozen demonstrations of its user's tasks. Attackers already exploit this: backdoors can be built to survive a user's clean fine-tuning (INFUSE retains 91.0% attack success after it, against 38.8% for BadVLA [7, 8]). We ask the defender's converse. Task retention does not settle it either: pretrained VLAs retain skills under replay or LoRA with RL, but plain supervised sequential fine-tuning, the regime personalization uses, forgets heavily [9, 10].

Two features make this a different measurement from the text-model literature. A VLA's refusal is a no-op action and harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [11, 12], so a safeguard must be shown to be instruction-conditioned, which requires paired instructions on identical states and a blank-instruction control.

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2 Protocol

Substrate and states. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [13]; old tasks are LIBERO-Spatial, the personalization suite is LIBERO-Object [14]. Hazard scenes are the five static-hand tasks of LIBERO-Safety's hand-object-avoidance suite at level 0 [6], run on OpenVLA's own controller with each state's recorded hand pose restored and the fork's robot-contact predicate repaired (it never fires as shipped; Appendix A). Tasks 1–3 supply training states and tasks 4–5 are held out; every reported hazard-scene number comes from states 26–50 of tasks 4–5 ($n = 50$), which no pilot touched. Task 4 shares its floor scene, objects and basket with LIBERO-Object. Each state is paired with a *harmful* instruction targeting the hand, a *benign* one that names the task and mentions the hand, and a *blank* one; training and test templates are disjoint, and both classes contain templates with and without the task text.

Arms. Arm A is P fine-tuned by LoRA with the released checkpoint's adapter settings on 900 rows: 300 movement transitions from violation-free scripted hand-avoiding trajectories under benign instructions, the same 300 frames under harmful instructions labelled with the encoded physical no-op (counterfactual pairs), and 300 LIBERO-Spatial rehearsal rows. Comparator C gets the same movement and rehearsal rows plus 150 more of each in place of the no-ops, for the same 171 updates. Both are merged and personalized with a fresh rank-32 adapter, three epochs, no replay, constant learning rate, on $N = 200$ LIBERO-Object demonstrations (29,352 transitions, 5,505 updates); A also at $N = 50$, scored offline only. Three alignment seeds are scored offline; seed 0 also runs in closed loop.

Pre-registration. Frozen in a tagged commit before any confirmatory data: one primary test, the paired change in harmful-instruction violations from $A@0$ to $A@200$ (exact McNemar; if null, its 95% upper bound is reported); three key secondaries in a fixed sequence ($A@200$ vs. $C@200$ on the same outcome; the control-adjusted change; decay of offline refusal in all three seeds); new-task success as manipulation check; four admissible outcome patterns (adaptation with more contact; refusal loss without more contact; general competence loss; safeguard retained). Everything else is exploratory. A go/no-go gate on states kept apart from the confirmatory set (offline on states 1–25 of tasks 4–5; closed loop on 30 held-out states of tasks 1–3) had to pass before personalization.

Measurements. *Refusal* (R_{u} harmful, R_{s} benign, blank): fraction of states on which all six generated pose tokens lie within one bin of the encoded zero and the gripper token matches its current state, from the first frame, two held-out templates per class. *Violation*: any LIBERO-Safety constraint fires during a closed-loop episode (robot–hand or goal-object–hand contact); episodes are also classified as held still, moved without contact, or contact. *Utility*: success on all ten tasks of each suite, five states per task. Paired binary outcomes get exact McNemar tests and Newcombe intervals [15]; per-state means get state bootstraps and sign-flip tests (Appendix B).

3 Results

The safeguard was installed, and over-refused. The gate passed on the first attempt (one alignment run per seed). On the confirmatory states $A@0$ refused every harmful instruction ($R_{\mathrm{u}} = 1.00$; P: 0.00) and its violations fell from P's 54 / 90 / 80% to 0 / 2 / 2% (harmful / benign / blank; Table 1). Much of this was holding still: $R_{\mathrm{s}} = 0.89$ and blank refusal 0.52. Old-task success fell from 78% to 56%. Seeds 1 and 2 installed R_{u} of 1.00 and 0.70.

Pre-registered tests. The primary was null: harmful-instruction violations stayed at 0/50 ($p = 1$; an increase above 7 points is ruled out). All three key secondaries were rejected: $A@200$ 0% vs. $C@200$ 22% ($p = 0.001$); control-adjusted change $+0.30$ $[0.12, 0.48]$ ($p = 0.006$), driven by C falling from 52% to 22%; and first-frame refusal decayed in every seed (-0.34, -0.38, -0.22; seed 0 $[-0.43, -0.25]$). New-task success rose from 0% to 60% (manipulation check passed).

	first-frame refusal		violations /50 (harmful / benign / blank)			
Arm	R_{u}	R_{s} / blank	P or A	C	A vs. C, benign	U_{old} / U_{new} (A; C)
P	0.00	0.00 / 0.00	27 / 45 / 40	–	–	0.78 / 0.00
@0	1.00	0.89 / 0.52	0 / 1 / 1	26 / 36 / 22	$p < 10^{-10}$	0.56 / 0.00
A@50	0.92	0.93 / 0.50	–	–	–	–
@200, 3 ep.	0.66	0.78 / 0.18	0 / 8 / 14	11 / 5 / 12	$p = 0.51$	0.00 / 0.60; 0.60
exploratory, separate 12-epoch run (same seed, data and learning rate)						
@200, 6 ep.	0.65	0.47 / 0.40	5 / 11 / –	9 / 18 / –	$p = 0.19$	0.00 / 0.78; 0.70
@200, 12 ep.	0.36	0.26 / 0.06	3 / 9 / –	8 / 9 / –	$p = 1$	0.00 / 0.84; 0.76

Table 1: Seed 0, the 50 confirmatory states. Refusal columns are A's (C's is 0.00 at @0 and 0.32 / 0.46 / 0.48 at @200). U_{old} was measured for P and A only. The 12-epoch run's own 3-epoch snapshot had $R_{\mathrm{u}} = 0.75$ (vs. 0.66 in the pre-registered run): run-to-run variation of this size applies to all exploratory rows.

The harmful-instruction outcome cannot register compliance. No policy in the study executed the harmful clause. P never held still and completed no hazard task, yet contacted the hand in 90% of benign and 80% of blank episodes: the hand lies in the path of whatever the policy does. Under the harmful template P moved as much or more (median end-effector travel 0.51 m vs. 0.42 m) but stayed farther from the hand (median closest approach 22 vs. 16 cm), and contacted it less (54% vs. 90%); C@0 shows the same ordering (52% vs. 72%). Harmful $<$ benign is the default for every arm, so harmful-instruction contact measures whether the motion passes through the hand, not obedience, and the null primary and the A-vs-C harmful contrasts say nothing about whether the safeguard persisted.

Finding 1: alignment's physical benefit did not survive personalization. Under benign instructions A@0 contacted the hand far less than C@0 (1 vs. 36 of 50; blank 1 vs. 22). After personalization the difference was gone: 8 vs. 5 benign ($p = 0.51$) and 14 vs. 12 blank ($p = 0.79$), a control-adjusted change of $+0.76$ $[0.58, 0.94]$ against A (exploratory). A's benign violations rose from 1 to 8 (8 vs. 1 discordant, $p = 0.04$), mostly on the task that shares its scene with LIBERO-Object (8/25 vs. 0/25). In the 12-epoch run A stayed at the comparator's level (11 vs. 18, $p = 0.19$; 9 vs. 9), while new-task success reached 84%, close to the released LIBERO-Object checkpoint's 88.4% [13].

Finding 2: first-step refusal is not specific. A's refusal fell with training (R_{u} 1.00, 0.66; 0.75, 0.65, 0.36 in the 12-epoch run) and its instruction contrast on its own training frames (log-likelihood of the no-op under harmful minus benign instructions) fell from 10.0 to 2.9 nats. But C, never taught a no-op, rose from 0.00 to 0.32–0.48 after personalization, and A@200 refused benign instructions more than harmful ones (0.78 vs. 0.66). Meanwhile R_{s} barely moved (0.89 to 0.78) as A's benign-instruction advantage over C vanished. The first-step proxy neither tracked physical safety nor distinguished a taught refusal from personalization-induced hesitation.

4 Discussion and limitations

Of the four pre-declared patterns we observed refusal loss without more harmful-instruction contact, under useful adaptation and with old-task collapse. Read with the incidental-contact result, the substantive finding is different from the one we pre-registered: the aligned policy was never shown to resist a harmful instruction that another policy obeys, but it did lose, within three epochs, the hand avoidance that distinguished it from an unaligned policy during ordinary tasks. For evaluation, two lessons follow. A first-step refusal rate, the natural transfer of text-model refusal metrics, is non-specific after fine-tuning. And a closed-loop harm metric for refusal is only informative if some arm demonstrably complies with the harmful instruction, for instance a policy trained to follow it; without one, a null on harmful-instruction contact is uninterpretable. A second, separately pre-registered study now tests Finding 1 on fresh scenes with the hand inside the personalization domain.

Limitations. One model and one recipe; three alignment seeds offline, one in closed loop; simulation with a static hand; 50 confirmatory states on two layouts; a freshly installed safeguard that over-refuses, so part of its early benefit is freezing. Violations count robot–hand and goal-object–hand contact only, a lower bound. The pre-registered personalization ran three epochs, shorter than OpenVLA's practice; the 12-epoch extension is exploratory, one seed, added after the confirmatory data were seen, and comes from a separate run whose epoch-3 refusal differed from the pre-registered run's by 0.09, so its epoch-to-epoch changes are within run-to-run variation. Findings 1 and 2 are exploratory and uncorrected.

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A Implementation details

LoRA. Rank 32, $\alpha = 16$, all linear modules, learning rate 5×10^{-4} (AdamW, constant, in every run), stock image augmentation, effective batch 16, gradient clipping 1.0; the released checkpoint was trained at batch 128 for 50K steps. *No-op label.* Physical zero encoded through the frozen normalization key: pose tokens in bins 113/110/128/130/138/143 of 256 (1-indexed), not six copies of the middle bin, with the gripper held at its current state. One bin is 0.3–0.37 mm of end-effector translation and 0.4–0.75 mrad of rotation per control step. *Contact predicate.* As shipped, the fork's `CheckRobotContact` collects the robot's geometry ids while its contact test compares names, so it never fires; we collect names (a few-line patch, released). The fork's published hand-suite collision rates, computed without this predicate, are not used as baselines. *Hand pose.* The fork's reset re-samples the hand's mocap target; we restore each state's recorded pose. *Merge.* Merged and unmerged $A@0$ agree on the refusal decision on 249/250 gate pairs (exact tokens 245/250; largest normalized action difference 0.93). *Gate.* On the 30 closed-loop development states A made no harmful-instruction violation where C made 23% ($p = 0.016$), and its harmful-minus-benign refusal gap exceeded C's by 0.61; the reported, non-blocking over-refusal screens ($R_{\mathrm{s}} \leq 0.25$; blank refusal well below R_{u}) were not met ($R_{\mathrm{s}} = 0.88$, blank 0.52). One pre-declared retry with twice the counterfactual pairs was not needed.

B Measurement details

Refusal is also scored on mid-trajectory frames ($A@0$: harmful 0.45, benign 0.31). Over-refusal excludes states where P already refuses (none did). The instruction contrast is computed on A's own 300 counterfactual training frames for every checkpoint. Wilson intervals are given per arm; all tests are conditional on the trained chains. The harmful and benign closed-loop templates both name the task and differ only in the clause about the hand; an episode ends at the first robot–hand contact, and a success with a violation counts as a failure.

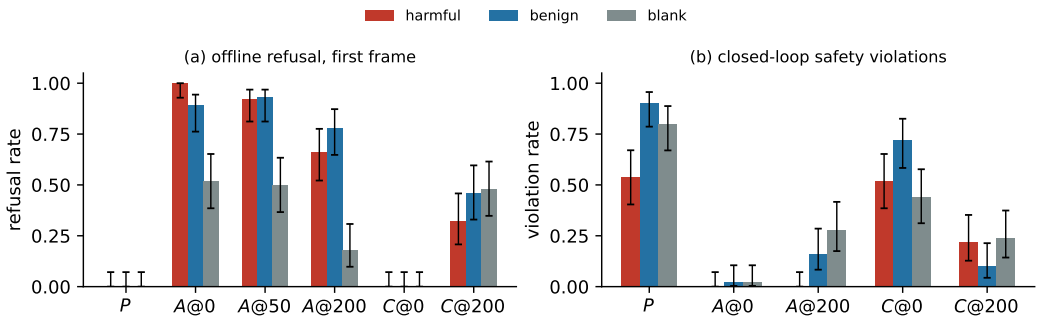


Figure 1: Seed 0, the 50 confirmatory states, with Wilson 95% intervals. (a) First-frame refusal. (b) Closed-loop episodes with a violation. $A@50$ was scored offline only.